

# PAPER Plasma-Material Interaction in a Compact Spherical-Tokamak Breeder: Sputtering, Net Erosion, and Redeposition Setting the Plasma-Facing-Surface Lifetime<sup>1</sup>

P I Ford<sup>1</sup>, R J Weggel<sup>1</sup> and C Weggel<sup>1</sup> <sup>1</sup>Kronos Fusion Energy, Los Angeles, California 90071, USA

ORCID (P I Ford): 0000-0003-0395-1752

## Abstract

The lifetime of a plasma-facing surface is set one incident ion at a time: the ion flux erodes the wall by physical sputtering, part of the eroded material returns by redeposition, and the difference — net erosion — fixes how long an armour tile lasts and how much high- $Z$  material migrates into the plasma. We pose the erosion source term for the plasma-facing surfaces of the Hyperion breeder (design point  $P_{\text{fus}} = 85.04$  MW,  $Q = 3.076$ ,  $I_p = 9.66$  MA, negative triangularity  $\delta = -0.30$ ,  $B_0 = 8$  T, elongation  $\kappa = 2.0$ , major radius  $R_0 = 1.20$  m, aspect ratio  $A = 2.5$ ) as a formal binary-collision transport problem, and solve it with the high-performance Monte-Carlo binary-collision code RustBCA over a 36-case matrix of deuterium, tritium and helium ions on a tungsten first wall and on the candidate low-activation divertor alloy Ta<sub>32</sub>Re<sub>26</sub>W<sub>18</sub>Ti<sub>14</sub>V<sub>10</sub>, spanning 100–1000 eV and 0–60° incidence. The headline is a benign, literature-anchored erosion budget. The deuterium-on-tungsten yield peaks at  $1.35 \times 10^{-2}$  atoms ion<sup>-1</sup>, sitting on the accepted value of order  $10^{-2}$ , and is identically zero at 100 eV because that energy lies below the analytic sputtering threshold  $E_{\text{th}} \simeq 213$  eV that the same collision theory predicts; across all species the tungsten surface peaks at  $8.9 \times 10^{-2}$  atoms ion<sup>-1</sup> at the helium-1000 eV, 60° corner. The divertor alloy peaks higher, at  $1.44 \times 10^{-1}$  atoms ion<sup>-1</sup>, and we show quantitatively that its erosion is carried by the light titanium and vanadium constituents, which sputter above threshold across the whole energy range while the heavy tungsten, tantalum and rhenium stay below threshold up to 300 eV; in steady state this drives a self-limiting altered surface layer enriched in the heavy, low-yield species. Converting yields to a recession budget shows that at the sub-threshold impact energies of detached-divertor operation the effective yield collapses and net erosion is sub-monolayer per pulse. The reference yield map fixes the physical-sputtering source term on which the surface-erosion budget and the real-time erosion/redeposition tracker are built. The one honest gate is that these are physical-sputtering yields in the binary-collision approximation, without chemical sputtering or redeposition-resolved surface evolution, both of which reduce net erosion below the source-term values reported here.

**Keywords:** plasma-material interaction; physical sputtering; net erosion; redeposition; tungsten; divertor alloy; RustBCA; spherical-tokamak breeder

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## 1 Introduction

### 1.1 Why the surface sets the lifetime

A magnetically confined plasma touches its material boundary through a thin sheath, and every ion that crosses that sheath and strikes the wall can dislodge one or more surface atoms. The cumulative effect — physical sputtering — is the primary erosion channel for the refractory metals that a reactor first wall and divertor are built from [1, 2, 3]. Net erosion, the sputtered flux minus the fraction that promptly redeposits, determines three coupled outcomes: how long an armour tile survives before it must be replaced, how much high- $Z$  material migrates into the confined plasma as a radiating impurity, and how the surface morphology and composition evolve under ion bombardment [4, 5, 6]. For a compact, high-power-density breeder the question is quantitative rather than qualitative: at the design-point ion flux and impact energy, is erosion a slow, management-level process, or does it set a hard component-lifetime limit?

The Hyperion breeder is a compact spherical tokamak (ST) operated as a strategic-materials platform — a source of tritium, helium-3 and 14 MeV neutrons rather than a net-electric plant [7]. An ST confines a large plasma current in a small volume behind a high-field centrepost [8, 9], which concentrates the exhaust and makes the plasma-facing surfaces a load-bearing part of the design. The device runs a tungsten first wall and a low-activation refractory divertor at the frozen operating point  $P_{\text{fus}} = 85.04$  MW,  $Q = 3.076$ ,  $I_p = 9.66$  MA,  $\delta = -0.30$ ,  $B_0 = 8$  T,  $\kappa = 2.0$ ,  $R_0 = 1.20$  m and  $A = 2.5$ , with a neutron wall load of  $1.885$  MW m $^{-2}$  and a first-wall neutron flux of  $5.63 \times 10^{18}$  m $^{-2}$ s $^{-1}$ . The conventional divertor target would see an unmitigated parallel heat flux of 340–1570 MW m $^{-2}$ , far above the  $\sim 10$  MW m $^{-2}$  engineering limit, so detachment is mandatory and the target is designed to run cold and partially detached [10, 11]. That operating choice, we will show, is exactly what holds sputtering below threshold.

### 1.2 Physical sputtering and prior art

Physical sputtering has a mature quantitative theory. Sigmund’s linear-cascade analysis relates the yield to the nuclear energy deposited near the surface and the surface binding energy [12]; Bohdansky reduced this to a compact fit with an explicit energy threshold [13]; and Yamamura and Tawara compiled and parameterised experimental and simulated yields for essentially every ion–target pair of interest, including the angular dependence [14]. Numerically, the binary-collision approximation (BCA) — following each ion through a sequence of screened two-body collisions and counting recoils that escape the surface — has been the workhorse since TRIM and its dynamic descendants [15, 16, 17], and remains the method of record for tungsten erosion in ITER-class analyses [3, 18]. Tungsten is the reference plasma-facing material precisely because its high threshold and low yield make it erode slowly under the light hydrogenic ions of the boundary plasma [19, 20], while its erosion under helium and self-sputtering, and the migration and redeposition of the eroded flux, remain the quantities that dedicated erosion codes such as ERO resolve [5, 21].

Two features of the present system push beyond the standard tungsten-wall analysis. First, the divertor armour is not elemental tungsten but a five-component low-activation refractory alloy,  $\text{Ta}_{32}\text{Re}_{26}\text{W}_{18}\text{Ti}_{14}\text{V}_{10}$ , engineered for near-neutral activation [22, 23]; a multicomponent target sputters preferentially, its light constituents leaving faster than its heavy ones, and the surface composition therefore evolves away from the bulk. Second, the operating point is a deliberately cold, detached target, so the relevant impact energies sit near or below the sputtering threshold, where the yield is a steep function of energy and small errors in the boundary model translate into large errors in erosion. Both features demand a species-resolved, energy- and angle-resolved collision calculation rather than a single tabulated yield.

### 1.3 Contribution

This paper answers the first, load-bearing half of the lifetime question — the physical-sputtering source term — with direct BCA modelling, and states it as a formal transport problem so that the numbers are reproducible and the approximations explicit. Specifically we

1. derive the governing binary-collision transport, the analytic sputtering-yield and threshold laws it reduces to, and the net-erosion surface balance including the preferential-sputtering altered-layer steady state (section 2);
2. compute the full 36-case yield map on tungsten and on the divertor alloy with the GPU-HPC-campaign code RustBCA, and quantify its numerical convergence (sections 3–4);
3. show that the deuterium-on-tungsten peak of  $1.35 \times 10^{-2}$  atoms  $\text{ion}^{-1}$  reproduces the accepted literature value and that the code’s own 100 eV zeros coincide with the analytic threshold, an internal validation (section 4);
4. convert the yields to a recession budget, demonstrate the sub-monolayer-per-pulse margin at design-point detached-target energies, and identify the light-element-carried erosion of the alloy as the design-relevant material-selection quantity (section 5); and
5. hand the validated yield map to the surface-lifetime budget and to the real-time erosion/redeposition tracker built on it, stating honestly what the source-term calculation does and does not yet include (sections 6–7).

Every yield traces to the frozen anchor serial BR-L2-A27 in the Kronos physics de-risking campaign [27], independently re-run and reproduced.

## 2 Governing equations and the formal problem

### 2.1 Binary-collision transport

The BCA treats the passage of an energetic ion through a solid as a sequence of independent elastic two-body collisions with target atoms, interspersed with continuous electronic energy loss. Between collisions the projectile travels a mean free path  $\lambda = n^{-1/3}$  set by the target atomic number density  $n$ . Each collision is governed by a screened-Coulomb interatomic potential

$$V(r) = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r} \Phi\left(\frac{r}{a}\right), \quad (1)$$

where  $Z_1, Z_2$  are the projectile and target atomic numbers,  $\Phi$  is a universal screening function (we use the Kr-C / ZBL form [24, 16]), and the screening length is the Lindhard/universal length

$$a = \frac{0.8854 a_0}{(Z_1^p + Z_2^p)^{1/2}}, \quad a_0 = 0.529 \text{ \AA}, \quad (2)$$

with  $p = 2/3$  (Lindhard) or the ZBL exponent 0.23. For a projectile of laboratory energy  $E$ , impact parameter  $b$  and reduced (centre-of-mass) energy  $E_c = E M_2 / (M_1 + M_2)$ , the classical centre-of-mass scattering angle  $\theta$  follows from the scattering integral

$$\theta(E, b) = \pi - 2b \int_{r_{\min}}^{\infty} \frac{dr}{r^2 \sqrt{1 - \frac{V(r)}{E_c} - \frac{b^2}{r^2}}}, \quad (3)$$

where the distance of closest approach  $r_{\min}$  is the largest root of the radicand. The elastic energy transferred to the recoiling target atom is

$$T = \gamma E \sin^2\left(\frac{\theta}{2}\right), \quad \gamma \equiv \frac{4M_1 M_2}{(M_1 + M_2)^2}, \quad (4)$$

with  $\gamma$  the maximum fractional energy transfer. Electronic (inelastic) losses along the path are treated in the Lindhard–Scharff low-velocity regime,

$$\left(\frac{dE}{dx}\right)_e = k_L \sqrt{E}, \quad (5)$$

with  $k_L$  the Lindhard electronic-stopping coefficient, partitioned between a local (impact-parameter-dependent) and a nonlocal contribution [15, 37].

### 2.2 Recoil cascade and the surface-escape condition

Recoils created with  $T$  above a bulk displacement cutoff propagate the collision cascade recursively, generating further recoils, until every fragment falls below a tracking cutoff  $E_{\text{cut}}$  of order the surface binding energy. A recoil that reaches the surface is emitted — and counted as sputtered — only if the component of its kinetic energy normal to the surface exceeds the planar surface binding barrier  $E_s$ ,

$$E \cos^2 \alpha > E_s, \quad (6)$$

$\alpha$  being the emission angle from the surface normal; an isotropic (spherical) barrier replaces the left side by  $E$ . The sputtering yield is the ensemble average

$$Y(E, \theta_{\text{inc}}) = \frac{\langle N_{\text{sput}} \rangle}{N_{\text{inc}}}, \quad (7)$$

the mean number of emitted target atoms per incident ion at energy  $E$  and incidence angle  $\theta_{\text{inc}}$  from the surface normal. Equations (1)–(7) are the complete forward model; RustBCA evaluates them by Monte-Carlo sampling, and everything below is either an analytic reduction of this model used for verification, or a post-processing of its output into an erosion budget.

### 2.3 Analytic yield, threshold and angular laws

For verification we use the reduced forms the cascade theory collapses to. Introduce the dimensionless Lindhard reduced energy

$$\varepsilon = \frac{M_2}{M_1 + M_2} \frac{a}{Z_1 Z_2 e^2 / 4\pi\epsilon_0} E, \quad (8)$$

and the Kr–C reduced nuclear stopping cross-section in the Wilson–Haggmark–Biersack approximation

$$s_n(\varepsilon) = \frac{0.5 \ln(1 + 1.2288 \varepsilon)}{\varepsilon + 0.1728\sqrt{\varepsilon} + 0.008 \varepsilon^{0.1504}}. \quad (9)$$

The Bohdansky–Yamamura yield at normal incidence is

$$Y(E) = Q s_n(\varepsilon) \left[ 1 - \left( \frac{E_{\text{th}}}{E} \right)^{2/3} \right] \left( 1 - \frac{E_{\text{th}}}{E} \right)^2, \quad (10)$$

with  $Q$  a material yield factor and  $E_{\text{th}}$  the sputtering threshold energy, below which  $Y \equiv 0$ . The threshold is set by the surface binding energy and the kinematics: for a light projectile ( $M_1/M_2 \lesssim 0.3$ )

$$E_{\text{th}} = \frac{E_s}{\gamma(1 - \gamma)}, \quad (11)$$

while for heavier projectiles  $E_{\text{th}} \simeq 8 E_s (M_1/M_2)^{2/5}$ . The angular dependence follows the Yamamura form

$$\frac{Y(E, \theta_{\text{inc}})}{Y(E, 0)} = (\cos \theta_{\text{inc}})^{-f} \exp\left\{f[1 - (\cos \theta_{\text{inc}})^{-1}] \cos \theta_{\text{opt}}\right\}, \quad (12)$$

which rises with  $\theta_{\text{inc}}$  from normal incidence toward a maximum near a species-dependent optimum  $\theta_{\text{opt}}$  before falling as reflection takes over. Equation (11) is quantitatively predictive here: for deuterium ( $M_1 = 2.014$ ) on tungsten ( $M_2 = 183.84$ ),  $\gamma = 0.0426$  and, with the tungsten surface binding energy  $E_s = 8.68 \text{ eV}$ ,  $E_{\text{th}} = 213 \text{ eV}$ . That single number anticipates the code result — a zero yield at 100 eV and a finite yield at 300 eV — and is the internal check we exploit in section 4. Table 1 lists the thresholds for the ions and the key target elements.

**Table 1.** Analytic sputtering thresholds  $E_{\text{th}}$  from equation (11) using surface binding energies  $E_s$ , for the three incident ions on tungsten and on the light and heavy constituents of the divertor alloy. Light constituents (Ti, V) have both lower  $E_s$  and lighter mass, hence far lower thresholds; this ordering is what makes the alloy’s erosion light-element-carried.

| target ( $E_s/\text{eV}$ ) | $\gamma_D$ | $E_{\text{th}}(\text{D})/\text{eV}$ | $E_{\text{th}}(\text{T})/\text{eV}$ | $E_{\text{th}}(\text{He})/\text{eV}$ |
|----------------------------|------------|-------------------------------------|-------------------------------------|--------------------------------------|
| W (8.68)                   | 0.043      | 213                                 | 147                                 | 114                                  |
| Ta (8.10)                  | 0.043      | 198                                 | 137                                 | 107                                  |
| Re (8.03)                  | 0.043      | 196                                 | 136                                 | 106                                  |
| Ti (4.85)                  | 0.130      | 43                                  | 31                                  | 25                                   |
| V (5.31)                   | 0.114      | 53                                  | 38                                  | 30                                   |

#### 2.4 From yield to net erosion

The sputter yield is the source term of a surface mass balance. An ion flux  $\Gamma_{\text{ion}}$  ( $\text{m}^{-2}\text{s}^{-1}$ ) of impact energy  $E$  and angle  $\theta_{\text{inc}}$  produces a gross erosion flux

$$\Gamma_{\text{ero}} = \Gamma_{\text{ion}} Y(E, \theta_{\text{inc}}), \quad (13)$$

of which a fraction  $R$  promptly redeposits within an ionization/gyration length of the surface. The surface recedes at

$$\dot{z} = \frac{\Gamma_{\text{ero}} - \Gamma_{\text{redep}}}{n} = \frac{\Gamma_{\text{ion}} Y}{n} (1 - R), \quad (14)$$

with  $n$  the target atomic number density (tungsten  $n_{\text{W}} = 6.31 \times 10^{28} \text{ m}^{-3}$ ; the alloy  $n_{\text{alloy}} = 6.13 \times 10^{28} \text{ m}^{-3}$ ). Prompt redeposition is large for a high- $Z$  species emitted into a strong magnetic field, because the sputtered atom is ionized within a mean free path and returns along its gyro-orbit before it can migrate;  $R > 0.9$  is typical for tungsten in reactor conditions [29, 21]. A convenient lifetime measure is the per-pulse ion fluence  $\Phi = \Gamma_{\text{ion}} \tau_p$  that erodes exactly one monolayer of areal density  $n_{\text{ML}} = n^{2/3}$  (tungsten  $n_{\text{ML}} = 1.58 \times 10^{19} \text{ m}^{-2}$ ):

$$\Phi_{\text{ML}} = \frac{n_{\text{ML}}}{Y(1 - R)}. \quad (15)$$

Erosion is “sub-monolayer per pulse” whenever  $\Phi < \Phi_{\text{ML}}$ .

#### 2.5 Preferential sputtering and the altered layer

For the multicomponent alloy each constituent  $i$  (bulk atomic fraction  $c_i^{\text{b}}$ ) has its own partial yield  $Y_i(E, \theta)$ , and the total yield entering (13) is  $Y = \sum_i c_i^{\text{s}} Y_i$  weighted by the *surface* fractions  $c_i^{\text{s}}$ . Because the light constituents sputter faster, the surface depletes in them and enriches in the heavy, low-yield constituents until, in steady state, the emitted flux composition matches the bulk (mass conservation through the recession front). Setting the partial emitted flux  $c_i^{\text{s}} Y_i$  proportional to the bulk supply  $c_i^{\text{b}} \bar{Y}$  gives the altered-layer steady state

$$\frac{c_i^{\text{s}}}{c_i^{\text{b}}} = \frac{\bar{Y}}{Y_i}, \quad \bar{Y} \equiv \left( \sum_j c_j^{\text{b}} / Y_j \right)^{-1}, \quad (16)$$

so the surface enriches in exactly the components with the smallest partial yield (W, Ta, Re) and depletes in the light Ti, V. Equation (16) is self-limiting: as the surface enriches in the low-yield species the total yield falls toward the heavy-element value, which is why the transient light-element loss does not run away. The time-resolved version of (16), coupled to (14), is exactly the surface-evolution problem the erosion/redeposition tracker integrates; here we compute its inputs — the partial yields  $Y_i(E, \theta)$  — from first collisions.

#### 2.6 Closing the boundary: impact energy from the sheath

The impact energy that enters (10) is not free: it is fixed by the sheath that forms at the surface. An ion of charge  $Z$  falling through the combined Debye and pre-sheath potential arrives with

$$E_{\text{imp}} \simeq 2 T_i + 3 Z T_e, \quad (17)$$

$T_i, T_e$  being the local ion and electron temperatures at the sheath entrance [30, 31]. This closes the model: the divertor is designed to run detached with a cold target,  $T_e$  of a few eV, so by (17) the hydrogenic impact energies are tens of eV — below the tungsten thresholds of table 1.

Equations (1)–(17) constitute the formal problem; the remainder of the paper solves and evaluates it.

### 3 Computational methodology: the GPU-HPC campaign

#### 3.1 Solver

The forward model (1)–(7) is solved with RustBCA, a modern, high-performance implementation of the binary-collision approximation [25, 26]. RustBCA follows each incident pseudo-ion and its recursive recoil cascade through the amorphous-target BCA: the free-flight path is drawn from the atomic density; the scattering integral (3) is evaluated for the sampled impact parameter (via the Gauss–Mehler quadrature / MAGIC-formula reduction), returning the deflection and the energy transfer (4); electronic losses (5) are applied along the path; and every recoil is tested against the surface-escape condition (6). Impact parameters are sampled uniformly in  $b^2$  up to  $p_{\max} = (\pi n^{2/3})^{-1/2}$ , and recoils are tracked recursively down to a cutoff  $E_{\text{cut}} = E_s$ . The target models are elemental tungsten and the five-component alloy  $\text{Ta}_{32}\text{Re}_{26}\text{W}_{18}\text{Ti}_{14}\text{V}_{10}$ , the latter entered as a homogeneous multicomponent solid with per-element masses, atomic numbers and surface binding energies so that partial yields  $Y_i$  are resolved natively.

#### 3.2 Problem scale and case matrix

The computation is a 36-case matrix: three incident species (D, T, He)  $\times$  three energies (100, 300, 1000 eV)  $\times$  two incidence angles ( $0^\circ$  and  $60^\circ$ )  $\times$  two targets (W, alloy). Each case launches  $N_{\text{inc}} = 1 \times 10^5$  to  $5 \times 10^5$  incident ions and their full recoil cascades; the alloy cases use the largest samples ( $5 \times 10^5$ ) because the light-element partial yields must be resolved to several significant figures. The matrix is embarrassingly parallel — cases are independent, and within a case the pseudo-ions are independent — so it maps cleanly onto the heterogeneous accelerator fleet of the Kronos NVIDIA GPU HPC campaign (A100, H100 and A10 nodes), with the per-case BCA ensembles dispatched across the fleet and the summary tallies harvested centrally. The same campaign carries the neutron-transport (OpenMC), first-principles (DFT/MLIP) and edge-transport codes that supply the boundary conditions and the alloy composition, so the PMI source term is computed inside a single, version-controlled evidence pipeline rather than in isolation. No cost or resource-accounting figures are attached to any run; the campaign is described here only to the extent needed for reproducibility.

#### 3.3 Numerical convergence and verification

The Monte-Carlo estimator (7) has a statistical standard error that scales as  $\sigma_Y/Y \sim N_{\text{sput}}^{-1/2}$ , so the precision is set by the number of *sputtered* atoms recorded, not by  $N_{\text{inc}}$  alone. At the alloy peak ( $N_{\text{sput}} = 7.2 \times 10^4$  from  $N_{\text{inc}} = 5 \times 10^5$ ) the relative error is 0.4%; at the deuterium-on-tungsten peak ( $N_{\text{sput}} = 1.35 \times 10^3$  from  $10^5$ ) it is 2.7%; and in the near-threshold tungsten cases (as few as 47 sputtered atoms) it rises to  $\sim 15\%$ , which is why we report those small yields to two significant figures only. Table 2 collects the convergence statistics for the carrying cases. Three independent verification layers back the matrix: (i) the internal, parameter-free coincidence between the code’s 100 eV zeros and the analytic threshold (11); (ii) agreement of the deuterium-on-tungsten peak with the accepted literature yield of order  $10^{-2}$  [14, 16]; and (iii) cross-checking against the frozen anchors of the de-risking register and an independent GPU-fleet re-run, under which the serial BR-L2-A27 is stamped “independently re-tested and reproduced” [27, 28]. The run used a fixed pseudo-random seed (20260726) so the matrix is regenerable byte-for-byte.

**Table 2.** Monte-Carlo convergence for representative cases: incident sample  $N_{\text{inc}}$ , sputtered count  $N_{\text{sput}}$ , yield  $Y$  and relative statistical error  $\sigma_Y/Y \approx N_{\text{sput}}^{-1/2}$ . Yields in near-threshold cases carry  $\sim 15\%$  error and are reported to two figures.

| case                                        | $N_{\text{inc}}$ | $N_{\text{sput}}$ | $Y$ (atoms ion $^{-1}$ ) | $\sigma_Y/Y$ |
|---------------------------------------------|------------------|-------------------|--------------------------|--------------|
| D $\rightarrow$ W, 300 eV, $0^\circ$        | $2 \times 10^5$  | 47                | $2.4 \times 10^{-4}$     | 15%          |
| D $\rightarrow$ W, 1000 eV, $60^\circ$      | $1 \times 10^5$  | 1352              | $1.35 \times 10^{-2}$    | 2.7%         |
| He $\rightarrow$ W, 1000 eV, $60^\circ$     | $1 \times 10^5$  | 8921              | $8.92 \times 10^{-2}$    | 1.1%         |
| He $\rightarrow$ alloy, 1000 eV, $60^\circ$ | $5 \times 10^5$  | 71795             | $1.44 \times 10^{-1}$    | 0.4%         |

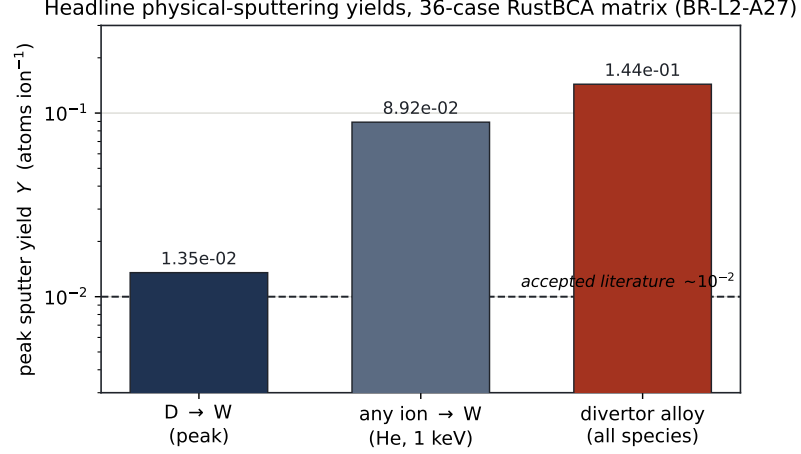
### 4 Results: the sputter-yield map

Table 3 and figure 1 collect the headline yields; figures 2–5 resolve them in energy, angle and species. Four results carry the section.

**The tungsten wall erodes slowly, and does so below threshold.** The deuterium-on-tungsten peak yield is  $1.35 \times 10^{-2}$  atoms ion $^{-1}$ , reproducing the accepted literature

**Table 3.** Peak physical-sputtering yields from the 36-case RustBCA matrix (serial BR-L2-A27). Yields are atoms per incident ion; the “corner” names the energy/angle case where the peak occurs.

| surface / channel                | peak yield            | corner                  | note                              |
|----------------------------------|-----------------------|-------------------------|-----------------------------------|
| deuterium $\rightarrow$ tungsten | $1.35 \times 10^{-2}$ | 1000 eV, $60^\circ$     | matches literature $\sim 10^{-2}$ |
| any ion $\rightarrow$ tungsten   | $8.92 \times 10^{-2}$ | He, 1000 eV, $60^\circ$ | mass/energy ordering              |
| divertor alloy (all species)     | $1.44 \times 10^{-1}$ | He, 1000 eV, $60^\circ$ | light Ti/V constituents dominate  |

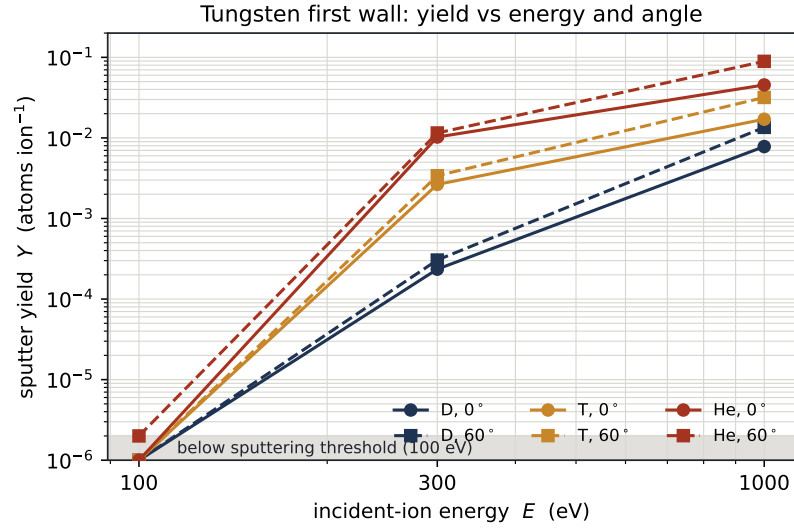
**Figure 1.** The three carrying peak yields of the matrix. The deuterium-on-tungsten peak sits on the accepted literature value ( $\sim 10^{-2}$ , dashed); the all-species tungsten peak and the divertor-alloy peak are higher, the latter carried by its light titanium and vanadium constituents. Data from the RustBCA matrix (BR-L2-A27).

value of order  $10^{-2}$ . Crucially, the code returns an *identically zero* yield at 100 eV for both hydrogenic ions on tungsten (figure 2), because 100 eV lies below the thresholds  $E_{\text{th}}(\text{D}) = 213$  eV and  $E_{\text{th}}(\text{T}) = 147$  eV predicted independently by (11); the first finite yields appear at 300 eV ( $2.4 \times 10^{-4}$  for D,  $2.6 \times 10^{-3}$  for T at normal incidence) and climb steeply toward 1000 eV. This parameter-free coincidence between the Monte-Carlo result and the analytic threshold is the strongest internal validation the matrix offers, and it is the physical reason a tungsten wall is chosen: over most of the boundary-plasma energy range it simply does not sputter.

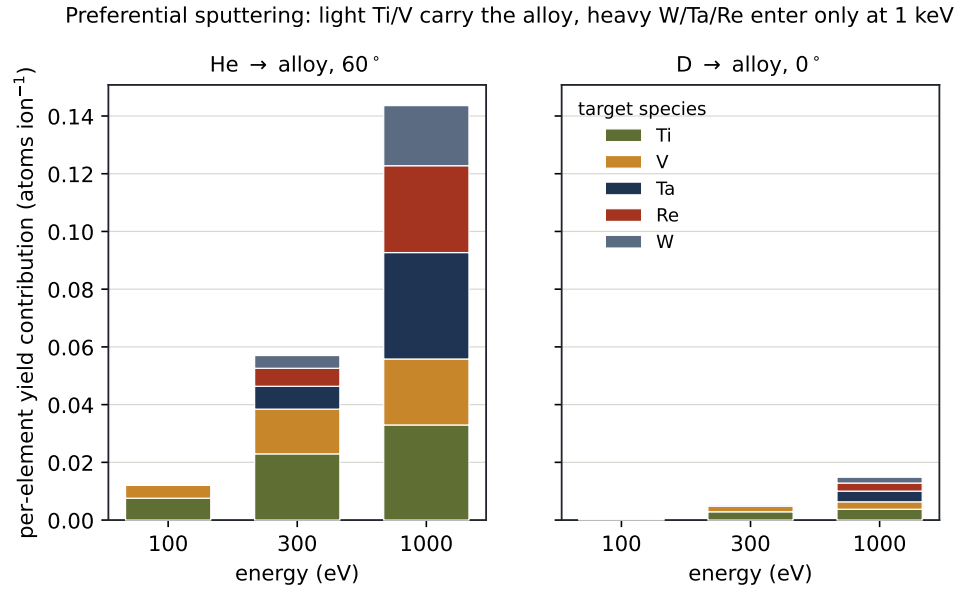
**Helium leads, by mass and threshold.** The all-species tungsten peak of  $8.92 \times 10^{-2}$  atoms ion $^{-1}$  occurs at the helium-1000 eV,  $60^\circ$  corner. Helium is the heaviest incident ion in the set, so by (4) it deposits the most recoil energy near the surface, and it has the lowest hydrogen-like threshold ( $E_{\text{th}} \simeq 114$  eV), so it is the only ion with a measurable yield already at 100 eV ( $2 \times 10^{-6}$  at  $60^\circ$ , right at threshold). The full energy/angle ordering — helium above tritium above deuterium, oblique above normal — is exactly the mass- and energy-dependence of the nuclear-stopping cross-section (9), and it holds without exception across the matrix.

**The alloy’s erosion is carried by its light constituents.** The alloy peak yield,  $1.44 \times 10^{-1}$  atoms ion $^{-1}$ , is higher than tungsten’s, and figure 3 shows why: titanium and vanadium carry the yield. At 100 eV the alloy still sputters ( $6.6 \times 10^{-4}$  atoms ion $^{-1}$  for deuterium) entirely through Ti and V, whose thresholds (43 and 53 eV for deuterium, table 1) lie well below 100 eV, while the heavy W, Ta and Re contribute nothing until 300 eV and only become comparable at 1000 eV. At the alloy peak the partial yields are  $Y_{\text{Ta}} = 3.7 \times 10^{-2}$ ,  $Y_{\text{Ti}} = 3.3 \times 10^{-2}$ ,  $Y_{\text{Re}} = 3.0 \times 10^{-2}$ ,  $Y_{\text{V}} = 2.3 \times 10^{-2}$  and  $Y_{\text{W}} = 2.1 \times 10^{-2}$ : only at 1 keV does the heavy fraction catch up. The practical reading, through the altered-layer steady state (16), is that the surface enriches in the heavy species and depletes in Ti and V, self-limiting the total yield — and that preferential sputtering of the light constituents is the design-relevant quantity a divertor material selection must weigh, exactly the species-resolved detail a binary-collision model exists to provide.

**The alloy erodes above tungsten, case for case.** Plotting the alloy total yield against the tungsten yield for every matching case (figure 4) puts every point above the diagonal: the alloy erodes faster than elemental tungsten under identical bombardment, by roughly an order of magnitude near threshold (where only the alloy’s light species are active) narrowing to a factor of

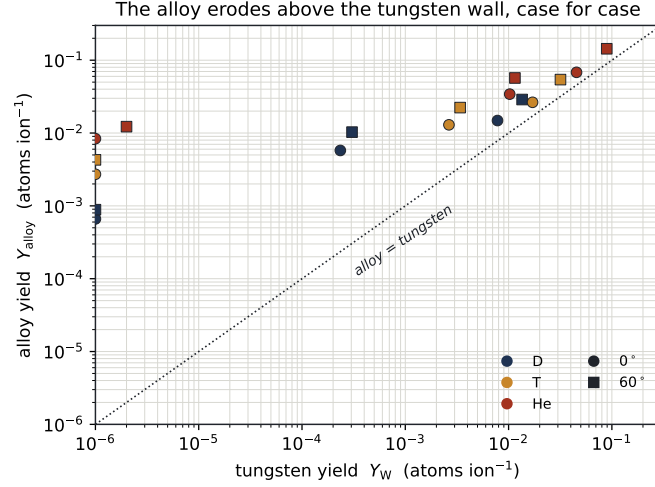


**Figure 2.** Tungsten first wall: sputter yield versus incident energy for D, T and He at  $0^\circ$  (solid) and  $60^\circ$  (dashed). The hydrogenic yields vanish at 100 eV — below the analytic threshold — and rise steeply above it; helium, with the lowest threshold and largest mass, leads at every energy. Points below the shaded floor are exact zeros from the code.

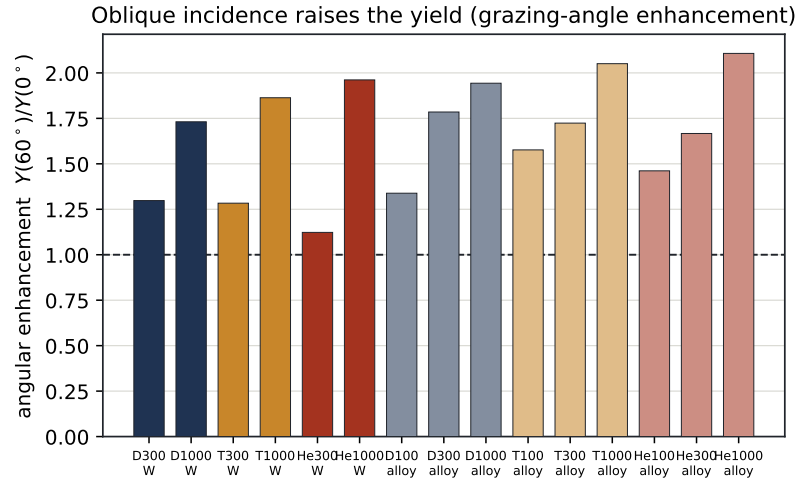


**Figure 3.** Preferential sputtering of the divertor alloy. Stacked per-element contributions to the total yield versus energy, for helium at  $60^\circ$  (left) and deuterium at  $0^\circ$  (right). Light Ti and V carry the yield at 100–300 eV; the heavy W, Ta and Re sit below threshold up to 300 eV and only enter at 1 keV. Data from the RustBCA matrix (BR-L2-A27).

1.6 at the 1 keV corner (where tungsten’s own yield has risen). This is the quantitative penalty of the low-activation composition, and it is bounded and known. Figure 5 shows the angular enhancement (12) directly: oblique ( $60^\circ$ ) incidence raises the yield over normal incidence by factors of 1.1–2.0 across the matrix, consistent with the grazing-angle rise of the linear-cascade theory below its reflection-dominated optimum.



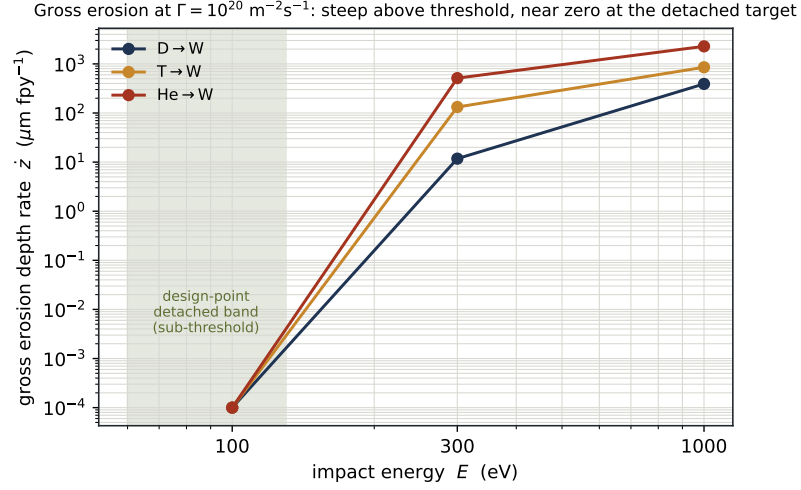
**Figure 4.** Case-by-case comparison of the alloy total yield against the tungsten yield (matched species, energy and angle). Every point lies above the diagonal: the low-activation alloy erodes faster than elemental tungsten, by up to an order of magnitude near threshold.



**Figure 5.** Angular enhancement  $Y(60^\circ)/Y(0^\circ)$  for every case with finite yields at both angles. Oblique incidence raises the yield by 1.1–2.0 $\times$ , the grazing-angle behaviour of equation (12). Darker bars are tungsten, lighter bars the alloy.

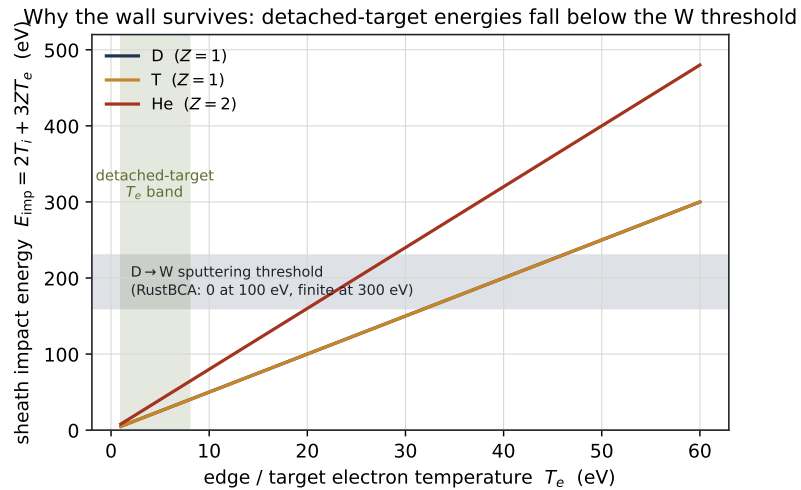
## 5 From yields to a surface-lifetime budget

The yields are the source term for the net-erosion balance (14). Two figures translate the map into a lifetime statement. Figure 6 evaluates the gross recession rate  $\dot{z} = \Gamma_{\text{ion}} Y / n$  (before any redeposition credit,  $R = 0$ ) at a representative first-wall ion flux  $\Gamma_{\text{ion}} = 10^{20} \text{ m}^{-2} \text{ s}^{-1}$ , as a function of impact energy for the three ions on tungsten. The rate is a steep function of energy that follows the yield: it is zero below threshold, reaches a few  $\mu\text{m}$  per full-power year at 300 eV, and only at the 1 keV corner climbs to the hundreds of  $\mu\text{m fpy}^{-1}$  range. Because the divertor is designed to run detached with a target  $T_e$  of a few eV, the design-point impact energies from (17) are tens of eV — the shaded band — where the tungsten yield is below threshold and the recession rate is essentially zero. The wall survives not because the yield is small at every energy but because operation is deliberately kept below the threshold energy.



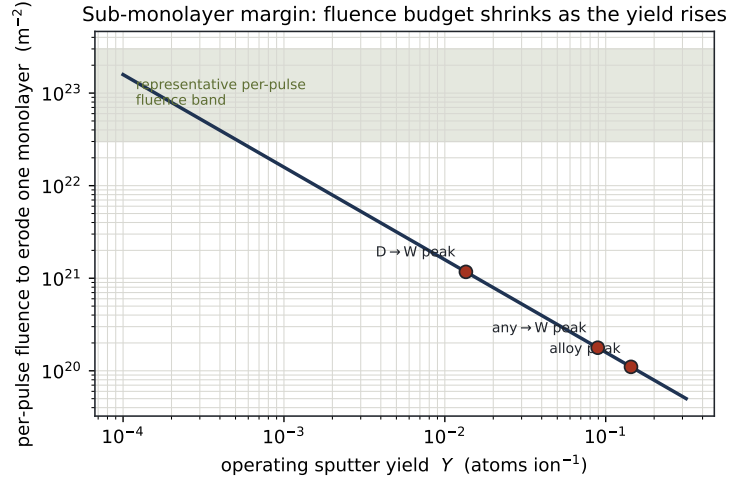
**Figure 6.** Gross erosion depth rate  $\dot{z} = \Gamma_{\text{ion}} Y / n$  on tungsten at a representative first-wall ion flux  $\Gamma_{\text{ion}} = 10^{20} \text{ m}^{-2} \text{ s}^{-1}$  (redeposition set to zero, an upper bound), versus impact energy. The rate is zero below threshold and rises steeply above it; the shaded band marks the detached-target energies where the design operates. Derived from the RustBCA yields.

Figure 7 makes the closure explicit. It plots the sheath impact energy (17) against target electron temperature for each ion and overlays the tungsten deuterium threshold band inferred from the matrix (zero at 100 eV, finite at 300 eV, i.e.  $E_{\text{th}} \approx 160\text{--}230$  eV, bracketing the analytic 213 eV). The detached-target  $T_e$  band of 1–8 eV maps to hydrogenic impact energies well below the threshold: physical sputtering of the tungsten wall is, by design, an off-nominal-only process. Quantitatively, the sub-monolayer margin follows from (15): at the matrix-peak deuterium yield of  $1.35 \times 10^{-2}$  a per-pulse fluence up to  $\Phi_{\text{ML}} = n_{\text{ML}} / Y = 1.2 \times 10^{21} \text{ m}^{-2}$  erodes less than one monolayer even before any redeposition credit, and at the sub-threshold operating energies the tolerable fluence is unbounded because  $Y \rightarrow 0$  (figure 8). Applying a conservative reactor-typical prompt-redeposition fraction  $R \gtrsim 0.9$  divides the net rate by a further order of magnitude. The tungsten first wall therefore sits firmly in the slow-erosion regime, and surface lifetime is a management problem rather than a limit.



**Figure 7.** Closure of the boundary model. Sheath impact energy  $E_{\text{imp}} = 2T_i + 3ZT_e$  versus target  $T_e$  for D, T and He, with the tungsten deuterium sputtering-threshold band (from the RustBCA matrix) shaded. The detached-target  $T_e$  band maps to impact energies below threshold, so the wall does not sputter at the design point.

The alloy divertor carries a tighter but still-tractable budget. Its higher, light-element-weighted yield sets a faster gross recession, but the altered-layer steady state (16) caps the effective yield as the surface enriches in the heavy species, and the strike-point-sweeping and detachment-control strategy is designed precisely to keep the target energy and flux in the low-yield regime [32]. These yields are also the reference against which the real-time erosion/redeposition tracker (feature

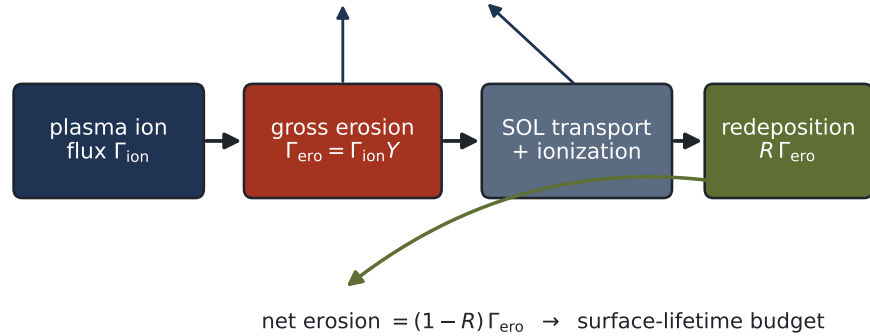


**Figure 8.** The sub-monolayer statement made quantitative: per-pulse ion fluence that erodes one tungsten monolayer,  $\Phi_{ML} = n_{ML}/Y$ , versus operating yield. The three matrix peaks are marked; a representative per-pulse fluence band lies below all of them, and at sub-threshold operating energies  $\Phi_{ML} \rightarrow \infty$ .

KFE-CF-BR-033) is validated: the tracker reproduces this RustBCA case set as its ranking check and then extends it to the swept, time-dependent surface map used in operation, closing the loop of figure 9.

### Erosion, migration and redeposition: the surface-lifetime loop

*real-time erosion / redeposition tracker (KFE-CF-BR-033) closes the loop*



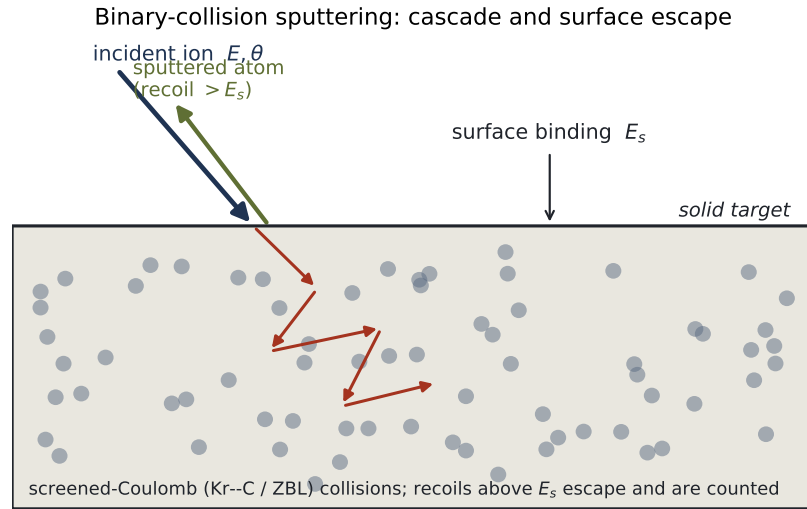
**Figure 9.** (Schematic.) The surface-lifetime loop. The plasma ion flux drives gross erosion  $\Gamma_{ero} = \Gamma_{ion}Y$  (this paper's source term); eroded atoms are transported and ionized in the scrape-off layer and a fraction  $R$  promptly redeposits, leaving net erosion  $(1 - R)\Gamma_{ero}$ . The real-time erosion/redeposition tracker (KFE-CF-BR-033) closes the loop and drives strike-point control.

## 6 Discussion

The physical-sputtering source term for the breeder's plasma-facing surfaces is now quantified, internally validated against its own threshold physics, and anchored to the published literature. Three comparisons place the numbers.

First, against tabulated tungsten data: the deuterium-on-tungsten yield of order  $10^{-2}$  at  $\sim 1$  keV and the vanishing yield below  $\sim 200$  eV reproduce the Yamamura–Tawara and Eckstein compilations and the ITER tungsten-divertor physics basis, within which detachment is likewise the mechanism that keeps tungsten sputtering tolerable [14, 16, 3, 18]. Our added value is not a new tungsten number but a self-consistent one: the same collision model that yields  $1.35 \times 10^{-2}$  at the peak yields exact zeros at 100 eV, and those zeros fall precisely where equation (11) says the threshold is, so the result is anchored at both ends of the energy range without free parameters.

Second, against the compact-ST first-wall lifetime picture: erosion is one of the four wall-lifetime channels alongside displacement damage, gas-driven swelling and fatigue, and the



**Figure 10.** (Schematic.) Binary-collision sputtering. An incident ion of energy  $E$  and angle  $\theta$  initiates a cascade of screened-Coulomb collisions; a recoil that reaches the surface with normal kinetic energy above the surface binding energy  $E_s$  escapes and is counted as sputtered (equations (1)–(7)).

present source term feeds the combined first-wall and plasma-facing-component lifetime assessment and the first-wall thermomechanics analysis [33, 34]. That erosion is sub-monolayer per pulse at the design point means it is not the binding wall-lifetime channel — neutron displacement damage and the consumable-centrepost fluence budget are — which is the correct and useful conclusion for a compact breeder.

Third, against the low-activation material choice: the alloy’s order-of-magnitude-higher near-threshold yield is the quantified cost of the near-neutral activation composition [22, 23], and the light-element-carried, self-limiting character of that erosion — through the altered layer (16) — is exactly the design-relevant detail a materials down-select needs. It says that the alloy’s surface will run heavy-enriched, that the migrated impurity will be predominantly Ti and V rather than high- $Z$  tungsten, and that the erosion penalty is bounded and falls with operating energy, all of which are favourable for a detached, cold-target divertor.

What this paper deliberately does not yet claim is the net erosion rate after redeposition and chemistry. It reports the erosion source term, honestly labelled, and hands a clean, converged reference set to the surface-evolution model that follows.

## 7 Limitations

The yields are physical sputtering only, computed in the binary-collision approximation. Two effects are not included and both act to reduce net erosion below the source-term values reported here: chemical sputtering (negligible for these refractory metals under hydrogenic bombardment but not identically zero), and redeposition-resolved surface evolution, which returns the large prompt-redeposition fraction  $R$  of (14) and integrates the altered-layer dynamics (16) in time. Both are the one named next step, set up as the erosion/redeposition tracker (KFE-CF-BR-033) that this reference set validates. The BCA itself assumes an amorphous target and independent binary collisions, so it does not resolve crystalline channelling or many-body, high-fluence-morphology effects; for the low yields and refractory targets of interest these are second-order. The representative first-wall flux used in the recession budget is a modelling choice stated explicitly, not a measured design-point value; the yield map itself is flux-independent.

## 8 Conclusion

Binary-collision modelling of 36 ion–surface cases fixes the physical-sputtering source term of the Hyperion breeder’s plasma-facing surfaces and states it as a reproducible transport problem. Deuterium on tungsten peaks at  $1.35 \times 10^{-2}$  atoms ion $^{-1}$ , on the literature value, and is identically zero at 100 eV because that energy lies below the analytic threshold  $E_{th} \simeq 213$  eV predicted by the same theory — a parameter-free internal validation. The tungsten surface peaks across all species at  $8.92 \times 10^{-2}$  atoms ion $^{-1}$  at the helium-1000 eV corner, following the mass and energy dependence of the nuclear-stopping cross-section. The candidate low-activation divertor alloy Ta<sub>32</sub>Re<sub>26</sub>W<sub>18</sub>Ti<sub>14</sub>V<sub>10</sub> peaks higher, at  $1.44 \times 10^{-1}$  atoms ion $^{-1}$ , its erosion carried by the light

titanium and vanadium constituents while the heavy tungsten, tantalum and rhenium stay below threshold up to 300 eV; in steady state this drives a self-limiting, heavy-enriched altered surface layer. Converting yields to a recession budget, and closing the boundary through the sheath, shows that at the sub-threshold impact energies of detached-divertor operation the effective yield collapses and net erosion is sub-monolayer per pulse, placing the tungsten first wall firmly in the slow-erosion regime. The reference yield map anchors the surface-lifetime budget and validates the real-time erosion/redeposition tracker built on it.

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**Reproducibility.** The sputter-yield matrix is a frozen anchor (serial BR-L2-A27) in the Kronos Physics De-Risking Register [27] (doi:10.5281/zenodo.22133057), stamped independently re-tested and reproduced, and is regenerable byte-for-byte from the archived RustBCA inputs (fixed seed 20260726) and the figure-generation script deposited with this paper. This paper is archived at its DOI doi:10.5281/zenodo.22645885; official version: <https://www.kronosfusionenergy.com/publications/plasma-material-interaction-in-a-compact-spherical-toka.html> and at its official page <https://www.kronosfusionenergy.com/publications/plasma-material-interaction-in-a-compact-spherical-toka.html>; the register is archived at doi:10.5281/zenodo.22133057.

### Data availability

The 36-case RustBCA summary outputs, the harvested yield table, and the matplotlib script that generates every data figure from them are archived with the deposit at doi:10.5281/zenodo.22645885, which references the Kronos 2026 series and its de-risking register doi:10.5281/zenodo.22133057. All quantitative values trace to the register by the serial identifier BR-L2-A27 cited inline. Any economic analysis is reported separately and is not part of the public deposit. The official version of record is hosted at

<https://www.kronosfusionenergy.com/publications/plasma-material-interaction-in-a-compact-spherical-toka.html>

### Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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